QUANTUM CRYPTOGRAPHY (BB84 PROTOCOL) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Quantum Cryptography (BB84 Protocol)

Question 1 10.0 points

Examine the Bloch sphere representation of the BB84 protocol quantum states in the diagram below. Describe the position of each of the four states ($|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$) in terms of their polar and azimuthal angles on the Bloch sphere using rectilinear and diagonal bases.

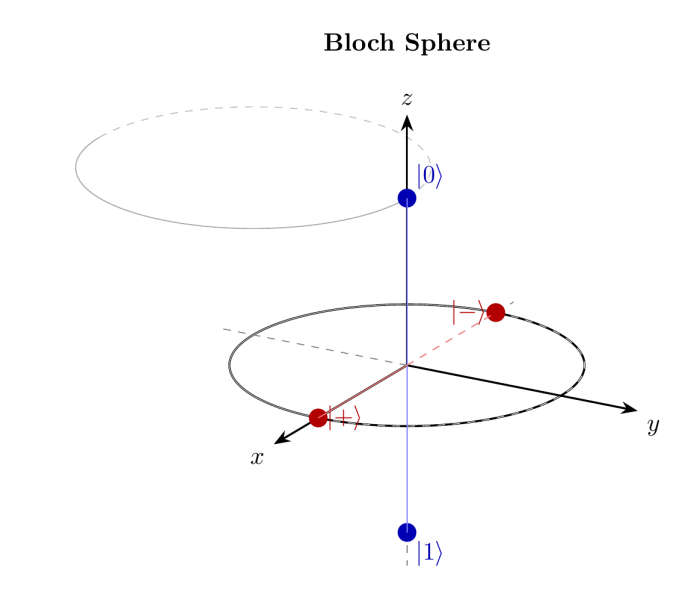

Fig. 1. A Bloch sphere diagram showing the positions of the four quantum states used in the BB84 protocol: |0⟩, |1⟩, |+⟩, and |-⟩. The Bloch sphere should be a 3D sphere with axes labeled x, y, and z. The state |0⟩ must be at the north pole (positive z-axis). The state |1⟩ must be at the south pole (negative z-axis). The state |+⟩ must be on the equator along the positive x-axis. The state |-⟩ must be on the equator along the negative x-axis. All four states must be clearly labeled. MISMATCH 1: The diagram shows the state labeled |+⟩ on the negative x-axis, but the DIAGRAM DESCRIPTION requires it to be on the positive x-axis (x=1). VISUAL FIX: The point and label for |+⟩ must be on the positive x-axis. MISMATCH 2: The diagram shows the state labeled |-⟩ on the positive x-axis, but the DIAGRAM DESCRIPTION requires it to be on the negative x-axis (x=-1). VISUAL FIX: The point and label for |-⟩ must be on the negative x-axis.

Question 2

10.0 points

Interpret the key exchange sequence in the diagram below for the BB84 protocol between Alice and Bob, including an intercept-resend attack by Eve. Explain how the protocol detects eavesdropping.

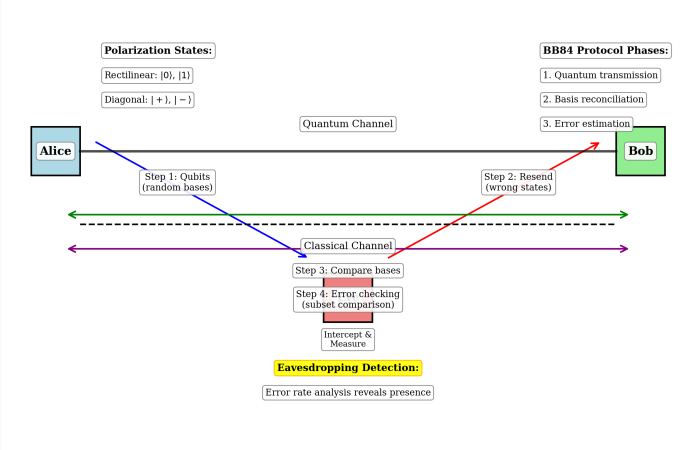

Fig. 2. A diagram illustrating the BB84 protocol sequence between Alice and Bob, including an intercept-resend attack by Eve. The diagram should show the following steps: 1. Alice sends a sequence of qubits encoded randomly in either the rectilinear or diagonal basis. 2. Eve intercepts the qubits, measures them in a randomly chosen basis, and resends them to Bob. 3. Bob receives the qubits and measures them in a randomly chosen basis. 4. Alice and Bob publicly compare a subset of their basis choices to detect eavesdropping. Include labeled arrows for each step, showing the flow of qubits and information between Alice, Eve, and Bob. Highlight the basis choices and potential discrepancies introduced by Eve's interference.

Question 3

10.0 points

Analyze the quantum key distribution channel in the diagram below, which illustrates the probability of eavesdropping detection versus sifted key error rate. Describe the trade-off between security and error rate.

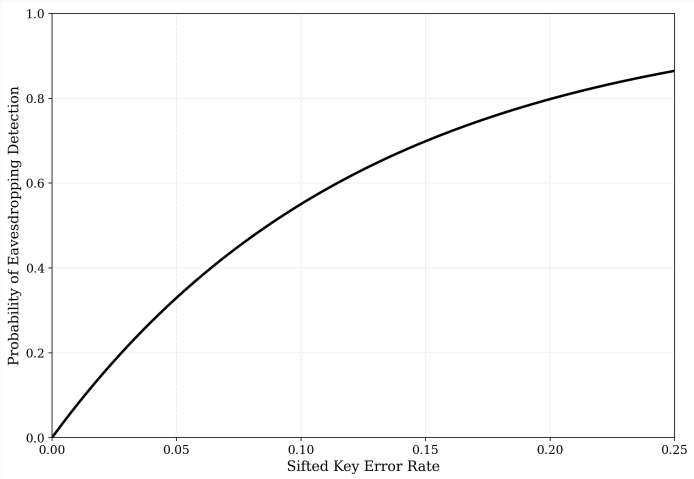

Fig. 3. A plot showing the probability of eavesdropping detection versus sifted key error rate in a quantum key distribution channel. The x-axis represents the sifted key error rate, and the y-axis represents the probability of eavesdropping detection. The curve should illustrate the trade-off between security (probability of detection) and error rate, highlighting how increasing the error rate affects the detection probability.

Question 4

10.0 points

Evaluate the quantum circuit shown in the diagram below that demonstrates the no-cloning theorem. Explain how this circuit prevents the cloning of an arbitrary quantum state.

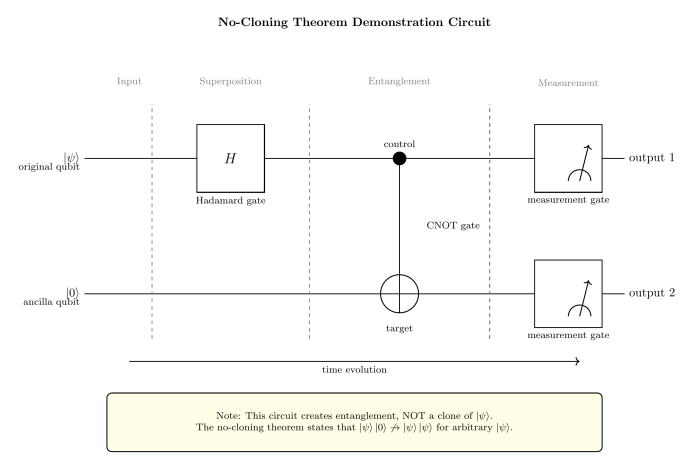

Fig. 4. A quantum circuit demonstrating the no-cloning theorem. The circuit includes two qubit lines: one for the original quantum state $|\psi\rangle$ and another for the ancilla qubit initialized to $|0\rangle$. The circuit has a CNOT gate with the original qubit as the control and the ancilla as the target. Additionally, there is a Hadamard gate on the original qubit before the CNOT gate. The circuit should clearly label the qubits and gates, showing the flow from the input state to the output, demonstrating the impossibility of cloning the quantum state.

Question 5

10.0 points

Examine the correlation curve between sifted key error rate and eavesdropping probability in the quantum key distribution session depicted in the diagram below. How can this curve guide secure communication practices?

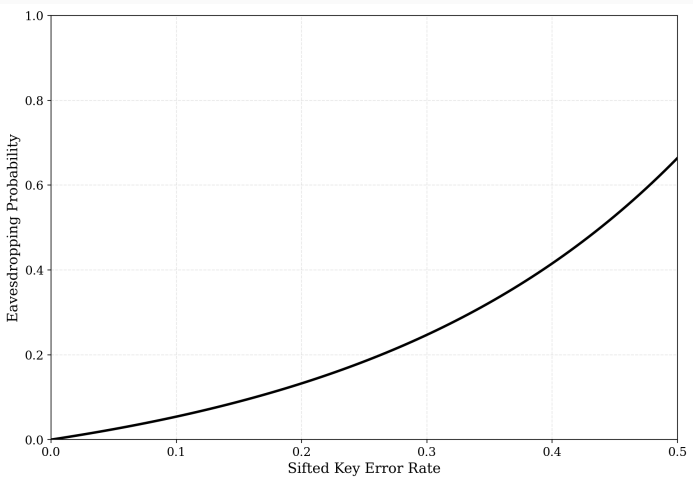

Fig. 5. A correlation curve showing the relationship between sifted key error rate and eavesdropping probability in a quantum key distribution session. The x-axis represents the sifted key error rate, and the y-axis represents the eavesdropping probability. The curve should illustrate how changes in the error rate affect the probability of eavesdropping, highlighting key points where the probability significantly increases or decreases. This visual aid helps in understanding how to maintain secure communication practices in quantum cryptography.

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